

1 Cross Norms

Notation 1.1. In this section, A_1, A_2 will denote Banach spaces. The norms on these spaces will be denoted by $\|\cdot\|$ unambiguously as we will distinguish between elements of A_1 and A_2 .

Notation 1.2. For Banach spaces A_1, A_2 , we will denote their *algebraic tensor product* by $A_1 \otimes A_2$. Its completion in some norm $\|\cdot\|_\beta$ will be denoted by $A_1 \otimes_\beta A_2$.

Definition 1.3. A norm $\|\cdot\|_c$ on $A_1 \otimes A_2$ is called a *cross norm* if it satisfies $\|x_1 \otimes x_2\|_c = \|x_1\| \|x_2\|$ for all $x_1 \in A_1, x_2 \in A_2$.

Definition 1.4. The *injective cross norm* $\|\cdot\|_\lambda$ on $A_1 \otimes A_2$ is defined by

$$\|\xi\| = \sup\left\{\left|\sum_{k=1}^n f(x_{1,k})g(x_{2,k})\right| : \begin{array}{l} \xi = \sum_{k=1}^n x_{1,k} \otimes x_{2,k} = \xi, \\ f \in A_1^*, \|f\| = 1, \\ g \in A_2^*, \|g\| = 1 \end{array} \right\}.$$

Remark 1.5. That $\|\cdot\|_\lambda$ is a cross norm is immediate from an application of the Hahn Banach Theorem.

Remark 1.6. Suppose a norm $\|\cdot\|_\beta$ satisfies $\|f \otimes g\|_{\beta^*} \leq \|f\| \|g\|$, then we also have

$$\left|\sum_{j=1}^m f(x_{1,j})g(x_{2,j})\right| \leq \|f \otimes g\|_{\beta^*} \left\|\sum_{j=1}^m x_{1,k} \otimes x_{2,j}\right\|_\beta \leq \left\|\sum_{j=1}^m x_{1,k} \otimes x_{2,j}\right\|_\beta$$

for f in A_1^* with $\|f\| \leq 1$ and $g \in A_2^*$ with $\|g\| \leq 1$.

Remark 1.7. As per the definitions in [1], all cross norms satisfy this condition so $\|\cdot\|_\beta$ is a cross norm if and only if $\|\cdot\|_\lambda \leq \|\cdot\|_\beta \leq \|\cdot\|_\gamma$.

Definition 1.8. The *projective cross norm* on $A_1 \otimes A_2$ is denoted by $\|\cdot\|_\gamma$ and defined as

$$\|\xi\| = \inf\left\{\sum_{k=1}^n \|x_{1,k}\| \|x_{2,k}\| : \xi = \sum_{k=1}^n x_{1,k} \otimes x_{2,k}\right\}.$$

Remark 1.9. Let $\|\cdot\|_\beta$ be a cross norm on $A_1 \otimes A_2$. For ξ in $A_1 \otimes A_2$, and $\epsilon > 0$, there exists a representation $\xi = \sum_{k=1}^n x_{1,k} \otimes x_{2,k}$ in $A_1 \otimes A_2$, such that

$$\|\xi\|_\beta \leq \sum_{k=1}^n \|x_{1,k} \otimes x_{2,k}\|_\beta = \sum_{k=1}^n \|x_{1,k}\| \|x_{2,k}\| \leq \|\xi\|_\gamma + \epsilon.$$

So, the projective norm majorises all cross norms. That it is itself a cross norm is evident from the computation

$$\|x_1\| \|x_2\| = \|x_1 \otimes x_2\|_\gamma \leq \|x_1 \otimes x_2\|_\lambda \leq \|x_1\| \|x_2\|.$$

The same computation reveals that all norms ‘between’ the two norms are cross norms. projective norm

1.1 C^* cross-norms

The C^* norms on $A_1 \otimes A_2$ can always be extended to unitisations.

Theorem 1.10. *Let A_1 and A_2 be C^* -algebras with a C^* norm $\|\cdot\|_\beta$ on $A_1 \otimes A_2$. There exists a C^* norm $\|\cdot\|_{\beta_0}$ on $(A_1)_I \otimes (A_2)_I$ which extends $\|\cdot\|_\beta$.*

Proof. Let π be a faithful representation of $A_1 \otimes_\beta A_2$. Then there exist representations π_i of A_i on the same Hilbert Space $\mathcal{H}$ satisfying

$$\pi(x_1 \otimes x_2) = \pi_1(x_1)\pi_2(x_2) = \pi(x_2)\pi_1(x_1)$$

for all x_1 in A_1 , all x_2 in A_2 and $\pi_1(x_1) = \lim_\lambda \pi(x_1 \otimes u_\lambda)$ for an approximate uni $\{u_\lambda\}$ of A_1 . Since π is faithful, $\|x_1 \otimes x_2\|_\beta = \|\pi(x_1 \otimes x_2)\|$. Define $\pi_i^0(x, \lambda) = \pi_i(x_i) + \lambda$ on $(A_i)_I$. Since the ranges of π_1^0 and π_2^0 commute with each other, $\pi_0(x_1 \otimes x_2) = \pi_1^0(x_1)\pi_2^0(x_2)$ defines a valid representation of $(A_1)_I \otimes (A_2)_I$. Define $\|x\|_{\beta_0} = \|\pi_0(x)\|$. This norm extends $\|\cdot\|_\beta$. $\square$

In general, the projective cross-norm is not a C^* norm [2, Remark 4.4]. So we consider

Definition 1.11. The maximal C^* norm defined by

$$\|x\|_{\max} = \sup\{\|\pi(x)\| : \pi \text{ is a representation of } A_1 \otimes A_2\}$$

The max norm is a cross norm.

Remark 1.12. For $\xi = \sum_{k=1}^n x_{1,k} \otimes x_{2,k}$, in $A_1 \otimes A_2$, we have

$$\|\pi(\xi)\| = \left\| \sum_{k=1}^n \pi(x_{1,k} \otimes x_{2,k}) \right\| \leq \sum_{k=1}^n \|\pi(x_{1,k} \otimes x_{2,k})\| \leq \sum_{k=1}^n \|x_{1,k} \otimes x_{2,k}\|.$$

This shows that $\|x\|_{\max} \leq \|x\|_\gamma$.

In addition, for any representation π_1 of A_1 and π_2 of A_2 , we have

$$\begin{aligned} \|\pi_1(x_1)\|\|\pi_2(x_2)\| &= \|\pi_1(x_1)\pi_2(x_2)\| = \|(\pi_1 \otimes \pi_2)(x_1 \otimes x_2)\| \\ &\leq \|x_1 \otimes x_2\|_{\max} \leq \|x_1 \otimes x_2\|_\gamma \leq \|x_1\|\|x_2\|. \end{aligned}$$

By choosing faithful representations π_1 and π_2 , the left hand side and right hand side become equal.

Definition 1.13. The minimal C^* cross-norm $\|\cdot\|_{\min}$ is defined by

$$\|\xi\|_{\min} = \sup\{\|(\pi_1 \otimes \pi_2)(\xi)\| : \begin{array}{l} \pi_1 \text{ is a representation of } A_1, \\ \pi_2 \text{ is a representation of } A_2 \end{array} \}.$$

It is immediate that $\|\cdot\|_{\min} \leq \|\cdot\|_{\max} \leq \|\cdot\|_\gamma$.

Remark 1.14. To see that $\|\cdot\|_{\min}$ is a cross norm, we only need to show that $\|\cdot\|_{\lambda} \leq \|\cdot\|_{\min}$. For f in A_1^* , there exist a representation π_1 of A_1 on a Hilbert Space $\mathcal{H}_1$ and ξ_1, η_1 in $\mathcal{H}_1$ satisfying $\|f\| = \|\xi_1\| \|\eta_1\|$ and $f(a) = \langle \eta_1, \pi_1(a) \xi_1 \rangle$ for all a in A_1 . Similarly, for g in A_2^* , there exist a representation π_2 of A_2 on a Hilbert space $\mathcal{H}_2$ and ξ_2, η_2 in $\mathcal{H}_2$ such that $g(b) = \langle \eta_2, \pi_2(b) \xi_2 \rangle$ for all b in A_2 and $\|g\| = \|\xi_2\| \|\eta_2\|$. This means for $x = \sum_{k=1}^n x_{1,k} \otimes x_{2,k}$ in $A_1 \otimes A_2$, we have

$$\begin{aligned}
& |(f \otimes g)x| \\
&= \left| \sum_{n=1}^k f(x_{1,k})g(x_{2,k}) \right| \\
&= \left| \sum_{k=1}^n \langle \eta_1, \pi_1(x_{1,k}) \xi_1 \rangle \langle \xi_2, \pi_2(x_{2,k}) \xi_2 \rangle \right| \\
&= \left| \sum_{k=1}^n \langle \eta_1, \pi_1(x_{1,k}) \xi_1 \rangle \langle \xi_2, \pi_2(x_{2,k}) \xi_2 \rangle \right| \\
&= \left| \sum_{n=1}^k \langle \eta_1 \otimes \eta_2, (\pi_1 \otimes \pi_2)(x_{1,k} \otimes x_{2,k}) (\xi_1 \otimes \xi_2) \rangle \right| \\
&= |\langle \eta_1 \otimes \eta_2, (\pi_1 \otimes \pi_2)(x) (\xi_1 \otimes \xi_2) \rangle| \\
&\leq \|\eta_1\| \|\eta_2\| \|(\pi_1 \otimes \pi_2)(x)\| \|\xi_1\| \|\xi_2\| \\
&= \|f\| \|g\| \|x\|_{\min}
\end{aligned}$$

In other words, $\|x\|_{\gamma} \leq \|x\|_{\min}$.

Thus, $\|\cdot\|_{\min}$ is a cross norm.

2 Universal Property

Theorem 2.1. For representations π_1 , respectively π_2 , of A_1 , respectively A_2 , onto a Hilbert Space $\mathcal{H}$ such that the images of π_1 and π_2 commute, there exists a unique representation π of $A_1 \otimes_{\max} A_2$ on the same space $\mathcal{H}$ such that

$$\pi(x_1 \otimes x_2) = \pi_1(x_1) \pi_2(x_2)$$

and the image of π is the C^* -algebra generated by the images of π_1 and π_2 .

Proof. Define $\pi(x_1 \otimes x_2) = \pi_1(x_1) \pi_2(x_2)$ for x_1 in A_1 , and x_2 in A_2 . For $x = \sum_{k=1}^n x_{1,k} \otimes x_{2,k}$ and $y = \sum_{l=1}^m y_{1,l} \otimes y_{2,l}$ in $A_1 \otimes A_2$, we have

$$\begin{aligned}
& \pi(x_{1,k} \otimes x_{2,k}) \pi(y_{1,l} \otimes y_{2,l}) \\
&= \pi_1(x_{1,k}) \pi_2(x_{2,k}) \pi_1(y_{1,l}) \pi_2(y_{2,l}) \\
&= \pi_1(x_{1,k}) \pi_1(y_{1,l}) \pi_2(x_{2,k}) \pi_2(y_{2,l}) \\
&= \pi_1(x_{1,k} y_{1,l}) \pi_2(x_{2,k} y_{2,l}) \\
&= \pi(x_{1,k} y_{2,l} \otimes x_{2,k} y_{2,l}) \\
&= \pi((x_{1,k} \otimes x_{2,k})(y_{1,l} \otimes y_{2,l}))
\end{aligned}$$

for every k in $\{1, 2, \dots, n\}$ and l in $\{1, 2, \dots, m\}$. Taking linear combinations over k, l we get

$$\pi(x)\pi(y) = \pi(xy).$$

Also, $\|(\pi_1 \otimes \pi_2)(\sum_{k=1}^n x_{1,k} \otimes x_{2,k})\| \leq \|\sum_{k=1}^n x_{1,k} \otimes x_{2,k}\|_{\max}$. So π is norm decreasing. Hence there exists an extension of π to all of $A_1 \otimes_{\max} A_2$.

Moreover, any other representation ρ which satisfies $\rho(x_1 \otimes x_2) = \pi_1(x_1) \otimes \pi_2(x_2)$ must agree with π on the completion because they agree on a dense subset. $\square$

Remark 2.2. This shows that $A_1 \otimes_{\max} A_2$ is embedded inside $A_1 \otimes_{\beta} A_2$ for every C^* cross-norm β .

Theorem 2.3. *Let A_1, A_2 be C^* -algebras and denote $A_1 \otimes_{\min} A_2$ by $A_{\min}$ and $A_1 \otimes_{\max} A_2$ by $A_{\max}$. Then*

1. *If $\omega_1 \in (A_1^*)_+$, and $\omega_2 \in (A_2^*)_+$, then $\omega = \omega_1 \otimes \omega_2$ is in $((A_{\max}^*)_+)$ and induces the representation $(\pi, \mathcal{H}, \xi)$ which is the tensor product of the representations $(\pi_1, \mathcal{H}_1, \xi_1)$ and $(\pi_2, \mathcal{H}_2, \xi_2)$.*

2.

$$\|x\|_{\min} = \sup \left\{ \frac{(\omega_1 \otimes \omega_2)(a^* x^* x a)^{1/2}}{(\omega_1 \otimes \omega_2)(a^* a)^{1/2}} : \begin{array}{l} \omega_1 \in (A_1^*)_+, \\ \omega_2 \in (A_2^*)_+, \\ a \in A_{\min} \end{array} \right\}$$

Proof. Define $\mathcal{H} = \mathcal{H}_1 \otimes \mathcal{H}_2$, $\xi = \xi_1 \otimes \xi_2$ and $\pi = \pi_1 \otimes \pi_2$. Then,

$$\begin{aligned} & \langle \xi, \pi(x_1 \otimes x_2) \xi \rangle \\ &= \langle \xi_1 \otimes \xi_2, (\pi_1(x_1) \otimes \pi_2(x_2))(\xi_1 \otimes \xi_2) \rangle \\ &= \omega_1(x_1) \omega_2(x_2) \\ &= (\omega_1 \otimes \omega_2)(x_1 \otimes x_2). \end{aligned}$$

Taking finite linear combinations, we get that $\omega(x) = \langle \xi, \pi(x) \xi \rangle$ for all x on the algebraic tensor product $A_1 \otimes A_2$. This shows that ω is positive in $A_{\max}$. Also, $\pi_1(A_1)\xi_1$ is dense in $\mathcal{H}_1$, $\pi_2(A_2)\xi_2$ is dense in $\mathcal{H}_2$. So their tensor product $(\pi_1(A_1)\xi_1) \otimes (\pi_2(A_2)\xi_2) = \pi(A_1 \otimes A_2)\xi$ is dense in $\mathcal{H}$. Thus, ξ is cyclic under π .

For the second assertion, pick x in $A_{\min}$. Given representations π_1 , respectively π_2 , of A_1 , respectively A_2 , we can express them as direct sums of irreducible representations as $\pi_1 = \oplus_i \pi_{1,i}$ and $\pi_2 = \oplus_j \pi_{2,j}$. Then

$$\|(\pi_1 \otimes \pi_2)(x)\| = \sup_{i,j} \{(\pi_{1,i} \otimes \pi_{2,j})(x)\}.$$

Since $\|\oplus_j a_j\| = \sup\{\|a_j\| : j\}$, we can rewrite

$$\|x\|_{\min} = \sup\{\|(\pi_1 \otimes \pi_2)(x)\| : \pi \text{ is a cyclic representation of } A_1 \otimes A_2.\}$$

Let ξ_i be the cyclic vector for π_i , $i = 1, 2$. Replacing π_1 , respectively π_2 with $\omega_1, \in (A_1^*)_+$, respectively $\omega_2 \in (A_2^*)_+$, we get

$$\begin{aligned}
& \|(\pi_1 \otimes \pi_2)(x)\| \\
&= \|(\pi_1 \otimes \pi_2)(x)^*(\pi_1 \otimes \pi_2)(x)\|^{1/2} \\
&= \sup\{\langle \eta, (\pi_1 \otimes \pi_2)(x^*x)\eta \rangle : \eta \neq 0\} \\
&= \sup\left\{\frac{\langle (\pi_1 \otimes \pi_2)(a)\xi, (\pi_1 \otimes \pi_2)(x^*x)(\pi_1 \otimes \pi_2)(a)\xi \rangle}{\langle (\pi_1 \otimes \pi_2)(a)\xi, (\pi_1 \otimes \pi_2)(a)\xi \rangle}\right\} \\
&= \sup\left\{\frac{\langle \xi, (\pi_1 \otimes \pi_2)(a^*x^*xa)\xi \rangle}{\langle \xi, (\pi_1 \otimes \pi_2)(a^*a)\xi \rangle} : a \neq 0, a \in A_1 \otimes A_2\right\} \\
&= \sup\left\{\frac{(\omega_1 \otimes \omega_2)(a^*x^*xa)}{(\omega_1 \otimes \omega_2)(x^*x)} : a \in A_1 \otimes A_2\right\}
\end{aligned}$$

□

3 Takesaki's Theorem

We will start this section with a lemma about product states, and consider a subset S_β of the continuous product pure states. Then we will prove that the injective cross norm and the injective cstar norm coincide if one of the factors is abelian. This proof makes use of an identification of $A_1 \otimes A_2$ with a subspace of the continuous A_1 valued functions on the character space of A_2 . This theorem will be useful in proving that the all cstar norms are equal if one of the factors is abelian. Finally we will show that the injective cstar norm is the minimal cstar norm, hence all cstar norms are cross norms.

Proposition 3.1. *If ω_1 is a positive linear functional on A_1 , and ω_2 is a positive linear functional on A_2 , then $\omega_1 \otimes \omega_2$ is positive on $A_1 \otimes A_2$.*

Proof. Suppose $x = \sum_{i=1}^n x_{1,i} \otimes x_{2,i}$ is in $A_1 \otimes A_2$. Then,

$$\begin{aligned}
& (\omega_1 \otimes \omega_2)(x^*x) \\
&= (\omega_1 \otimes \omega_2)\left(\sum_{i,j=1}^n (x_{1,i} \otimes x_{2,i})^* \otimes (x_{1,i} \otimes x_{2,i})\right) \\
&= (\omega_1 \otimes \omega_2)\left(\sum_{i,j=1}^n (x_{1,i}^* x_{1,i} \otimes x_{2,i}^* x_{2,i})\right) \\
&= \sum_{i,j=1}^n \omega_1(x_{1,i}^* x_{1,i}) \omega_2(x_{2,i}^* x_{2,i}).
\end{aligned}$$

For complex numbers $\lambda_1, \dots, \lambda_n$, we have

$$\begin{aligned}
& \sum_{i,j=1}^n \overline{\lambda_i} \lambda_j \omega_2(x_{2,i}^* x_{2,j}) \\
&= \omega_2\left(\sum_{i,j=1}^n \overline{\lambda_i} \lambda_j x_{2,i}^* x_{2,j}\right) \\
&= \omega_2\left(\left(\sum_{i=1}^n \lambda_i x_{2,i}\right)^* \left(\sum_{j=1}^n \lambda_j x_{2,j}\right)\right) \geq 0.
\end{aligned}$$

This means that the matrix $A = [\omega_2(x_{2,i}^* x_{2,j})]$ is positive semidefinite and has a representation $A = \sum_{l=1}^n [\overline{r_{l,i}} r_{l,j}]$, where $[r_{l,i}]$ is a scaling of some rank one projection onto the subspace spanned by $(r_{l,1}, \dots, r_{l,n})$. This means

$$\begin{aligned}
& (\omega_1 \otimes \omega_2)(x^* x) \\
&= \sum_{i,j=1}^n \omega_1(x_{1,i}^* x_{2,j}) \sum_{l=1}^n \overline{r_{l,i}} r_{l,j} \\
&= \sum_{l=1}^n \omega_1\left(\sum_{i,j=1}^n \overline{r_{l,i}} r_{l,j} x_{1,i}^* x_{2,j}\right) \\
&= \sum_{l=1}^n \omega_1\left(\left(\sum_{i=1}^n r_{l,i} x_{1,i}\right)^* \left(\sum_{j=1}^n r_{l,j} x_{2,j}\right)\right) \geq 0.
\end{aligned}$$

□

Notation 3.2. The set of all Pure states on A_1 is denoted by $P(A_1)$.

Notation 3.3. Given a cstar norm β on $A_1 \otimes A_2$, we denote

$$S_\beta = \{(\omega_1 \otimes \omega_2) : \omega_1 \otimes \omega_2 \text{ is continuous in the } \beta \text{ norm}.\}$$

Lemma 3.4. *The set S_β is a closed subset of $P(A_1) \times P(A_2)$ with the product weak* topology.*

Proof. Let $\omega_{1,\alpha} \otimes \omega_{2,\alpha}$ be a net in S_β such that $\lim_\alpha \omega_{1,\alpha}(x_1) = \omega_1(x_1)$ for each x_1 in A_1 , and $\lim_\alpha \omega_{2,\alpha}(x_2) = \omega_2(x_2)$ for each x_2 in A_2 . For a fixed $x =$

$\sum_{l=1}^k x_{1,l} \otimes x_{2,l}$ in $A_1 \otimes A_2$,

$$\begin{aligned}
\omega_1 \otimes \omega_2(x) &= \omega_1 \otimes \omega_2\left(\sum_{l=1}^k x_{1,l} \otimes x_{2,l}\right) \\
&= \sum_{l=1}^k \omega_1(x_{1,l})\omega_2(x_{2,l}) \\
&= \sum_{l=1}^k \lim_i \omega_{1,\alpha}(x_{1,l})\omega_{2,\alpha} \lim_i(x_{2,l}) \\
&= \sum_{l=1}^k \lim_i (\omega_{1,\alpha} \otimes \omega_{2,\alpha})(x_{1,l} \otimes x_{2,l}) \\
&= \lim_{\alpha} (\omega_{1,\alpha} \otimes \omega_{2,\alpha})(x)
\end{aligned}$$

For x in $A_1 \otimes A_2$, we have $|(\omega_{1,\alpha} \otimes \omega_{2,\alpha})(x)| \leq \|x\|_{\beta}$ for each i . So, $|\omega_1 \otimes \omega_2(x)| \leq \|x\|_{\beta}$. This means $\omega_1 \otimes \omega_2$ is also in S_{β} . $\square$

Theorem 3.5. *If A_2 is abelian, then the min norm and the λ norm coincide.*

Proof. Let A_2 be $C(\Omega)$ for some compact Hausdorff Space Ω . For

$$x = \sum_{i=1}^n x_{1,i} \otimes x_{2,i}$$

in $A_1 \otimes A_2$, define $\Lambda(x): \Omega \rightarrow A_1$ by

$$\Lambda(x)(\omega) = \sum_{i=1}^n x_{2,i}(\omega)x_{1,i}.$$

The space $C(\Omega, A_1)$ of continuous functions taking values in A_1 is equipped with the C^* -algebra structure

$$\begin{aligned}
\|a\| &= \sup\{|a(\omega)|: \omega \in \Omega\} \\
(\lambda a + \mu b)(\omega) &= \lambda a(\omega) + \mu b(\omega) \\
(ab)(\omega) &= a(\omega)b(\omega) \\
a^*(\omega) &= (a(\omega))^*
\end{aligned}$$

for a, b in $C(\Omega, A_1)$, λ, μ in $\mathbb{C}$ and ω in Ω . For $x = \sum_{i=1}^n x_{1,i} \otimes x_{2,i}$ in $A_1 \otimes A_2$,

we have

$$\begin{aligned}
\|\Lambda(x)\| &= \sup\{\|\sum_{i=1}^n x_{2,i}(\omega)x_{1,i}\| : \omega \in \Omega\} \\
&= \sup\{|\sum_{i=1}^n x_{2,i}(\omega)\phi(x_{1,i})| : \phi \in A_1^*, \|\phi\| \leq 1, \omega \in \Omega\} \\
&= \sup\{\|\Lambda(\sum_{i=1}^n \phi(x_{1,i})x_{2,i})\| : \phi \in A_1^*, \|\phi\| \leq 1, \omega \in \Omega\} \\
&= \sup\{|\sum_{i=1}^n \phi(x_{1,i})\psi(x_{2,i})| : \phi \in A_1^*, \|\phi\| \leq 1\} \\
&= \sup\{|\phi \otimes \psi(\sum_{i=1}^n x_{1,i} \otimes x_{2,i})| : \phi \in A_1^*, \|\phi\| \leq 1, \psi \in A_2^*, \|\psi\| \leq 1\} \\
&= \|x\|_\lambda.
\end{aligned}$$

Regard each ω as a one dimensional representation of A_2 and let π_1 be a faithful representation of A_1 . Then

$$\begin{aligned}
&\|x\|_\Lambda \\
&= \sup\{\|\sum_{i=1}^n x_{2,i}(\omega)x_{1,i}\| : \omega \in \Omega\} \\
&= \sup\{\|\pi_1(\sum_{i=1}^n x_{2,i}(\omega)x_{1,i})\| : \omega \in \Omega\} \\
&= \sup\{\|(\pi_1 \otimes \pi_\omega)(\sum_{i=1}^n x_{1,i} \otimes x_{2,i})\| : \omega \in \Omega\} \\
&= \|(\pi_1 \otimes \oplus_{\omega \in \Omega} \pi_\omega)(\sum_{i=1}^n x_{1,i} \otimes x_{2,i})\| \\
&= \|x\|_{\min}.
\end{aligned}$$

□

We now move on to prove that the injective cstar cross norm agrees with all cstar norms if one of the factors is abelian.

Lemma 3.6. *Let $(\omega_1 \otimes \omega_2)$ be in S_β , u , respectively v , be a unitary in A_1 , respectively A_2 . Then $(u^*\omega_1 u \otimes v^*\omega_2 v)$ is also in S_β .*

Proof. Let x be in $A_1 \otimes A_2$. Then we have

$$\begin{aligned}
&|(u^*\omega_1 u \otimes v^*\omega_2 v)(x)| \\
&= |(\omega_1 \otimes \omega_2)((u \otimes v)x(u^* \otimes v^*))| \\
&= \|(\omega_1 \otimes \omega_2)\|_{\beta^*} \|((u \otimes v)x(u^* \otimes v^*))\|_\beta \\
&= 1\|x\|_\beta
\end{aligned}$$

because $\|\omega_1 \otimes \omega_2\|_{\beta*} = (\omega_1 \otimes \omega_2)(1) = 1$. □

Theorem 3.7. *Let A_2 be abelian. Then $\beta = \lambda$.*

Proof. Suppose ω is in $P(A_\beta)$. Then ω is of the form $\omega = \omega_1 \otimes \omega_2$. Then,

$$\begin{aligned}
& \|x\|_\beta \\
&= \sup\{|\omega(x)| : \omega \in P(A_\beta)\} \\
&= \sup\{|\omega_1 \otimes \omega_2(x)| : \omega_1 \in P(A_1) \omega_2 \in P(A_2)\} \\
&= \sup\{\|\pi_{\omega_1} \otimes \pi_{\omega_2}(x)\| : \omega_1 \in P(A_1) \omega_2 \in P(A_2)\} \\
&\leq \sup\{\|\pi_1 \otimes \pi_2(x)\| : \begin{array}{l} \pi_1 \text{ is some representation of } A_1 \\ \pi_2 \text{ is some representation of } A_2 \end{array} \} \\
&= \|x\|_{\min}.
\end{aligned}$$

Now we will prove that S_β is all of $P(A_1) \times P(A_2)$. Towards contradiction, suppose this is not the case. Then there are open sets U_1 , respectively U_2 of $P(A_1)$, respectively $P(A_2)$ such that $U_1 \times U_2$ does not intersect S_β . Replacing U_1 by

$$\bigcup_{u \in \mathcal{U}(A_1)} uU_1u^*$$

and U_2 by

$$\bigcup_{v \in \mathcal{U}(A_2)} vU_2v^*,$$

we get that U_1 and U_2 are unitarily invariant and their product still does not intersect S_β , by Lemma 3.6. This means $K_1 = U_1^c$ and $K_2 = U_2^c$ are closed subsets of $P(A_1)$ and $P(A_2)$ which satisfy $S_\beta \subseteq K_1 \times P(A_2) \cup P(A_1) \times K_2$. Since K_1 is closed,

$$I_{K_1} = \{x_1 \in A_1 : \omega(x_1) = 0 \text{ for all } \omega \in K_1\}$$

is a closed ideal of A_1 . This is due to [3, Theorem 4.15]. This means there exists a nonzero positive element y_1 in I_{K_1} such that $\omega_1(y_1) = 0$ for all ω_1 in K_1 . Similarly, there exist a nonzero y_2 with the corresponding property. So $(\omega_1 \otimes \omega_2)(y_1 \otimes y_2) = 0$ for all $\omega_1 \otimes \omega_2$ in $P(A_\beta)$. Since all pure states in $P(A_\beta)$ are of this form, this would mean $y_1 \otimes y_2 = 0$. So, for x in $A_1 \otimes A_2$, we have

$$\begin{aligned}
& \|x\|_\lambda = \|x\|_{\min} \\
&= \sup\{\|(\pi_1 \otimes \pi_2)(x)\| : \begin{array}{l} \pi_1 \text{ is some irreducible representation of } A_1 \\ \pi_2 \text{ is some irreducible representation of } A_2 \end{array} \} \\
&\leq \|x\|_\beta.
\end{aligned}$$

□

A Representations

Theorem A.1. *Show that a $*$ -homomorphism $\pi: M_n(A) \rightarrow B(\mathcal{H})$ can be expressed as $\pi([a_{i,j}]) = [\rho(a_{i,j})]$ where $\rho: A \rightarrow B(\mathcal{K})$ is a $*$ -homomorphism of A on the Hilbert Space $\mathcal{K}$ which satisfies $\mathcal{H} = \oplus_{j=1}^n \mathcal{K}$.*

Proof. ... □

Multiplication is well defined.

B Questions for later

Questions I thought of while reading the text, scheduled to be answered later.

What is uniform closure [3, proof of Prop 4.13 page 211]?

Why are we interested in S_β ? Is $\omega_1 \otimes \omega_2$ even positive?

Is the weak star topology weaker than weak topology?

What is a nondegenerate representation? Is the image dense or closed?

If u , respectively v , is a unitary in A_1 , respectively A_2 , then is $\|(u \otimes v)x(u^* \otimes v^*)\| = \|x\|$?

References

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